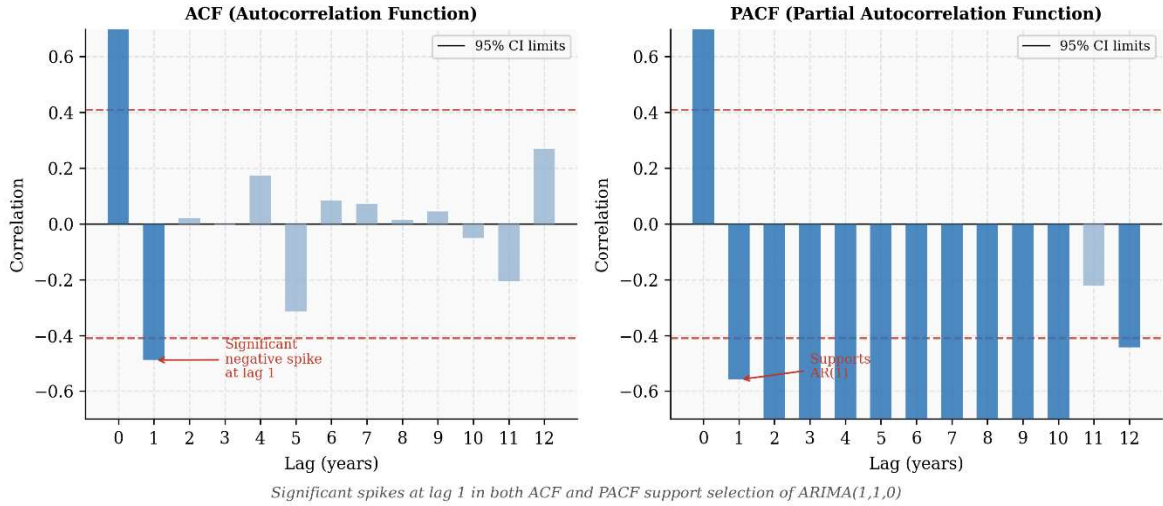




CF and PACF of First-Differenced Yearly NDVI Series. Dashed lines = 95% confidence limits ($\pm 1.96/\sqrt{n}$). Significant lag-1 spikes support ARIMA(1,1,0) selection.

Figure 4.4: ACF and PACF Plots of First-Differenced Yearly NDVI Series

The ACF plot of the differenced series shows a significant negative spike at lag 1 falling outside the 95% confidence bands, while all higher lags remain within the bands. This pattern is characteristic of a first-order moving average process, suggesting an MA(1) component. The PACF plot also shows a significant spike at lag 1, with subsequent lags falling within the confidence limits. This pattern is consistent with a first-order autoregressive process, suggesting an AR (1) component. On the basis of these patterns, candidate ARIMA models with orders $p \in \{0, 1\}$ and $q \in \{0, 1\}$, combined with $d = 1$, were considered for formal evaluation.

Autocorrelation and partial autocorrelation analysis were performed using the `acf()` and `pacf()` functions on the first-differenced NDVI series. Complete implementation details are provided in Appendix A, Section A.2.

4.2.9 ARIMA Model Selection

Three candidate ARIMA models were fitted to the first-differenced yearly NDVI series and compared using the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Root Mean Square Error (RMSE). Lower AIC and BIC values indicate a better

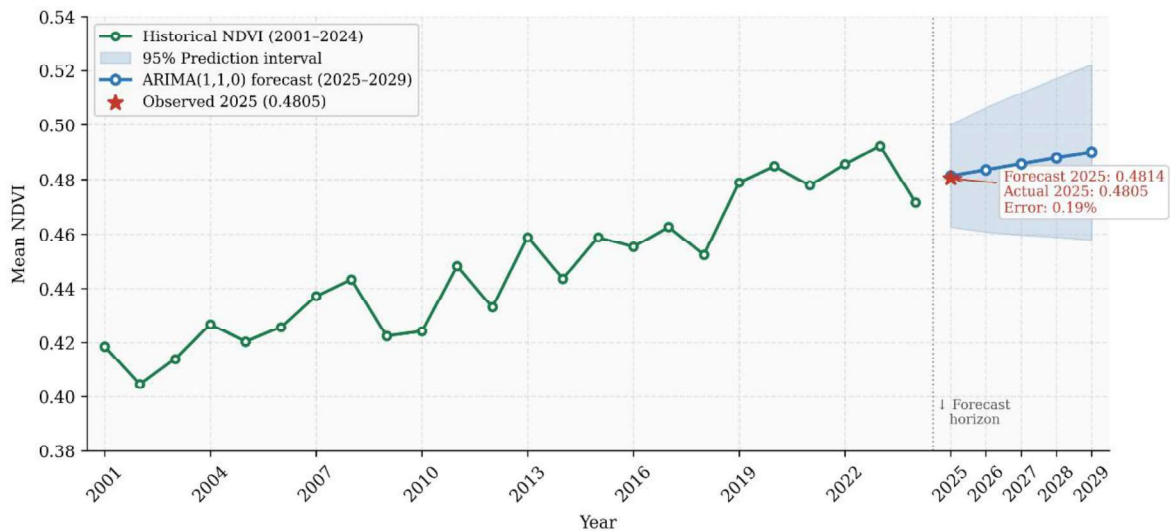
noise — that is, they show no pattern, no autocorrelation, and approximate normality — the model is considered adequate.

The residual time plot shows that residuals fluctuate randomly around zero without any discernible trend or periodicity, indicating that the systematic component of the series has been adequately captured. The ACF plot of residuals shows that all autocorrelation values lie within the 95% confidence limits, confirming the absence of significant serial dependence in the residuals. The histogram of residuals is approximately bell-shaped and centered near zero, supporting approximate normality. The Ljung-Box test produced a test statistic of $Q^*(12)$ with a p-value of 0.158, which is greater than 0.05. The null hypothesis of no residual autocorrelation is therefore not rejected, confirming that the ARIMA(1,1,0) model residuals satisfy the white noise assumption and the model is suitable for forecasting.

Residual diagnostic analysis was performed using residual plots, autocorrelation analysis, histogram inspection, Q–Q normality assessment, and the `Box.test()` function in R. Detailed implementation is provided in Appendix A, Section A.2.

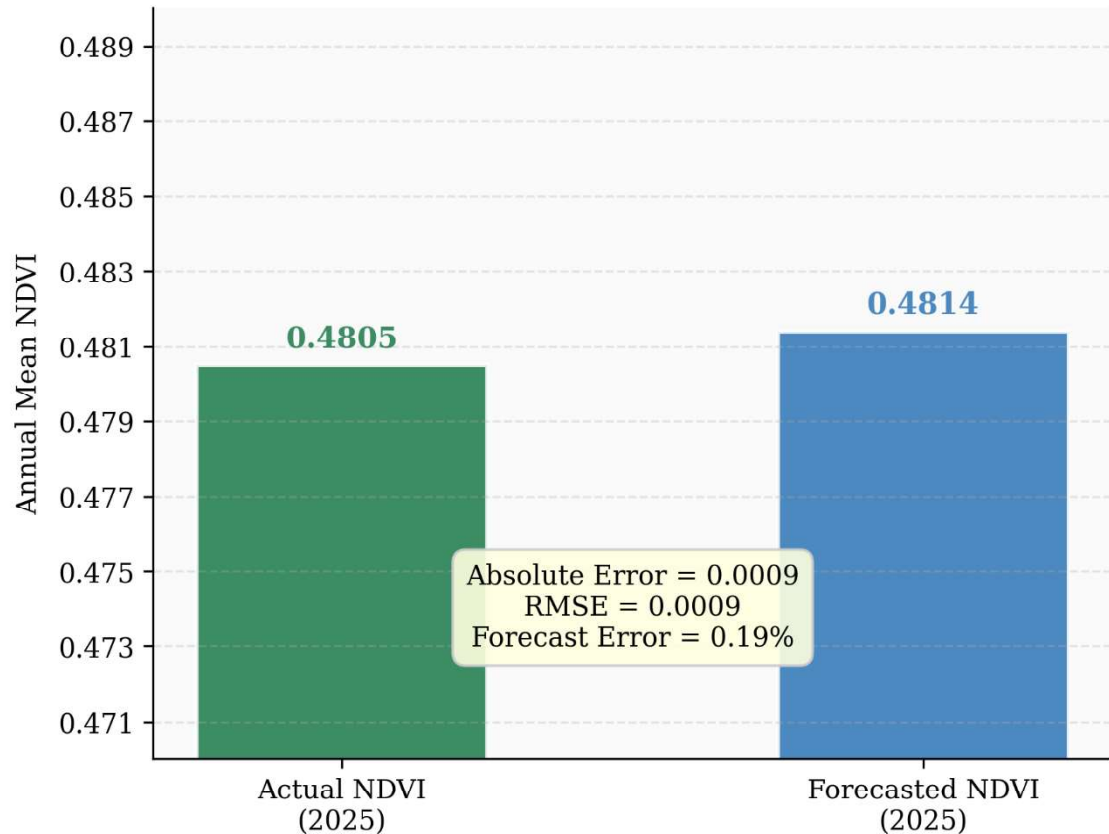
4.2.11 Yearly NDVI Forecast (2025–2029)

Following model confirmation, the ARIMA(1,1,0) model was used to generate point forecasts and 95% prediction intervals for the five-year period from 2025 to 2029.



Annual NDVI Forecast for Aligarh District (2025–2029) using ARIMA(1,1,0). Blue line = point forecast; shaded band = 95% prediction interval. Red point = observed 2025 value (0.4805). Forecast error = 0.19%.

Figure 4.5: Projected Annual Mean NDVI for Aligarh District (2025–2029)



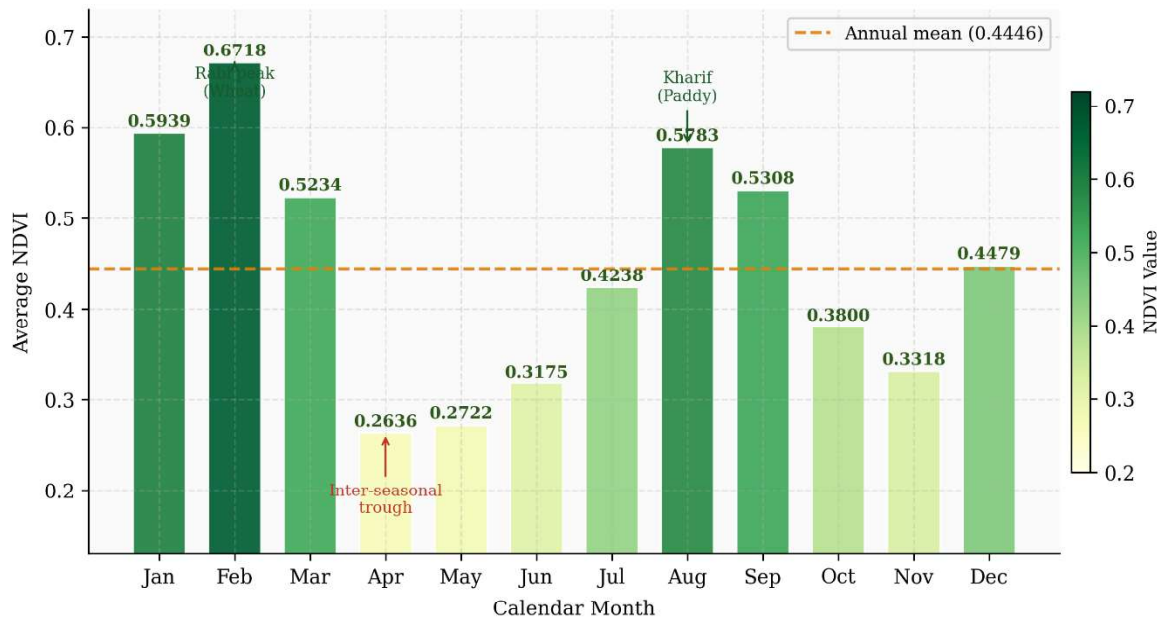
Actual versus Forecasted Annual Mean NDVI for Aligarh District (2025). Absolute Error = 0.0009; RMSE = 0.0009; Percentage Error = 0.19%.

Figure 4.6: Actual versus Forecasted Annual Mean NDVI for Aligarh District (2025)

The comparison in Table 4.6 and Figure 4.6 shows that the ARIMA(1,1,0) model achieved a forecast error of only 0.0009 NDVI units, corresponding to a percentage error of 0.19%. This level of accuracy is considered very satisfactory for an environmental time series of this type. The forecasted and observed values are virtually indistinguishable in both the numerical comparison and the graphical display. The RMSE for this single-year validation is 0.0009, well below typical thresholds considered acceptable for satellite-derived NDVI forecasting. These results confirm that the model reliably captures the long-term trajectory of annual vegetation conditions in Aligarh district and is suitable for short-term prediction.

level shows a gradual upward shift — consistent with the long-term trend identified in the yearly analysis. No abrupt structural breaks or anomalous years are apparent in the series.

4.3.4 Seasonal Pattern Analysis



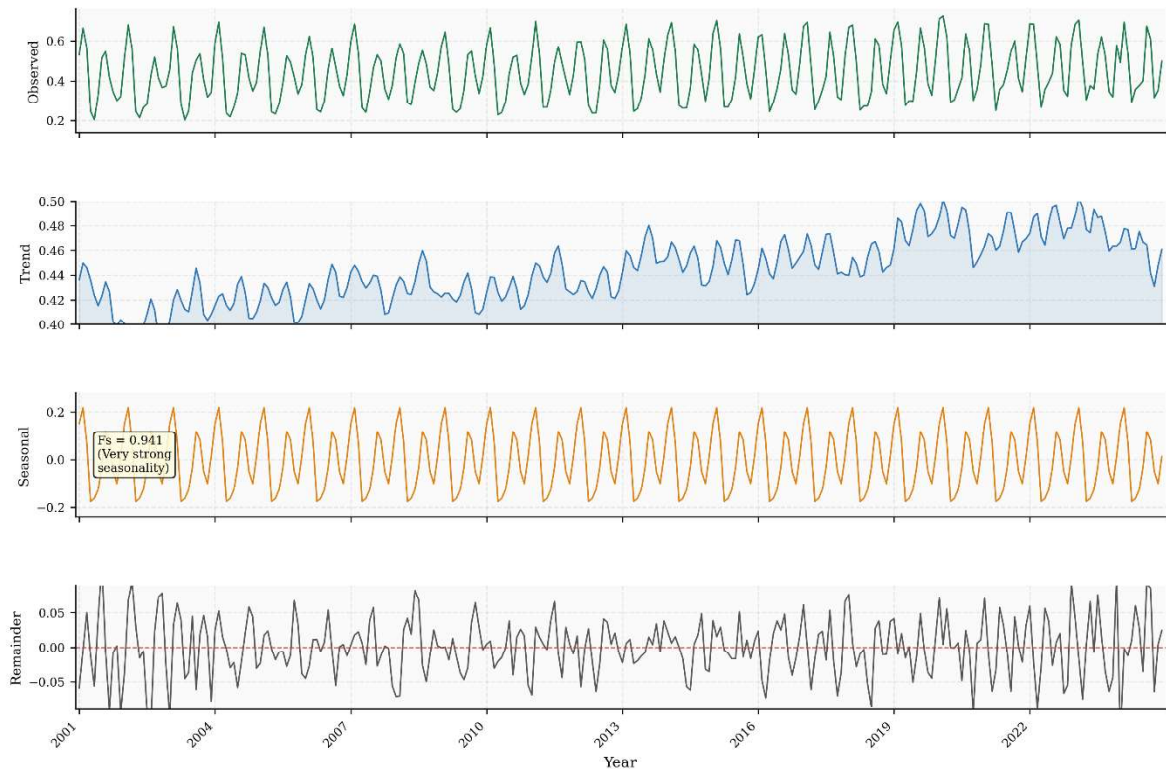
Average Monthly NDVI for Aligarh District (2001–2024). Colour intensity reflects NDVI magnitude. February = Rabi peak (0.6718); April–May = inter-seasonal trough (0.26–0.27); August = Kharif peak (0.5783).

Figure 4.8: Average Monthly NDVI Pattern for Aligarh District (2001–2024) Bimodal agricultural seasonal cycle

Figure 4.8 displays the average NDVI for each calendar month computed across all 24 years. The plot confirms the two-season agricultural pattern of Aligarh. The Rabi season peak centered on February dominates the annual cycle, with mean NDVI values exceeding 0.67. The inter-seasonal trough in April and May, when NDVI drops to approximately 0.26–0.27, marks the period between Rabi harvest and Kharif sowing. The Kharif season produces a secondary peak in August at approximately 0.58, following monsoon establishment in June–July.

4.3.5 STL Decomposition and Seasonal Strength

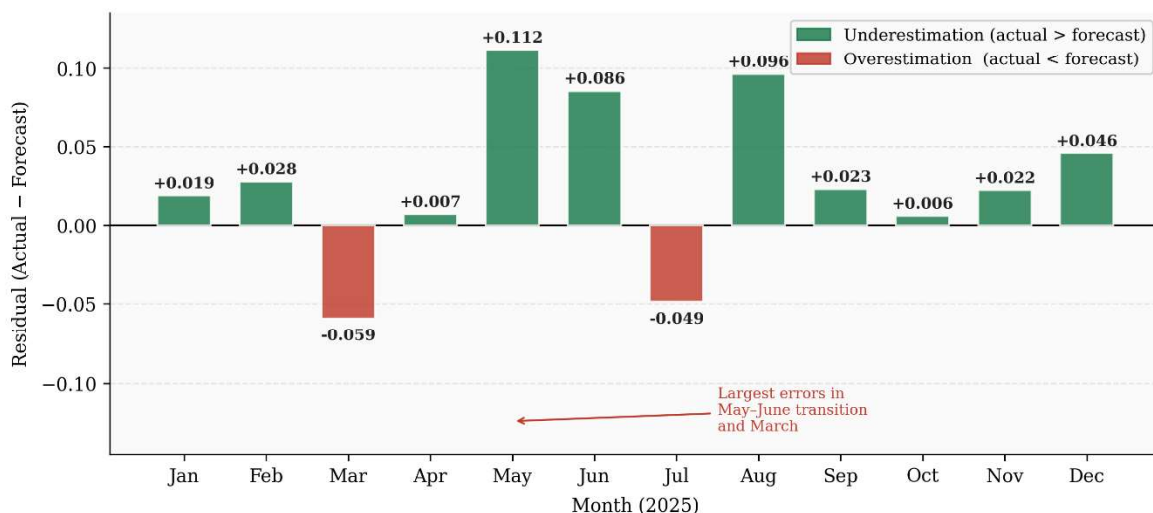
Seasonal and Trend decomposition using Loess (STL) was applied to the monthly NDVI series to separate the trend component, the seasonal component, and the remainder component. STL is well suited to this dataset because it can handle both the regular seasonal pattern and the gradual long-term trend without imposing rigid distributional assumptions.



STL Decomposition of Monthly NDVI Series for Aligarh District (2001–2024). Seasonal strength $F_s = 0.941$, confirming very strong seasonality. Trend shows gradual increase from ~ 0.42 to ~ 0.48 over the study period.

Figure 4.9: STL Decomposition of Monthly NDVI Series — Aligarh District (2001–2024)

Figure 4.9 presents the four-panel STL output. The observed series (top panel) reproduces the regular oscillation pattern seen in Figure 4.6. The trend component (second panel) shows a smooth, gradually increasing trajectory from approximately 0.420 in 2001 to approximately 0.478 by 2024, consistent with the upward trend identified in the yearly analysis. The seasonal component (third panel) shows a stable, highly regular pattern that repeats almost



Monthly Forecast Residuals (Actual – Forecast) for Aligarh District (2025). Positive = underestimation; negative = overestimation. Largest errors in May–June (inter-seasonal transition period). No persistent directional bias.

Figure 4.12: Monthly Forecast Residuals - Aligarh District (2025)

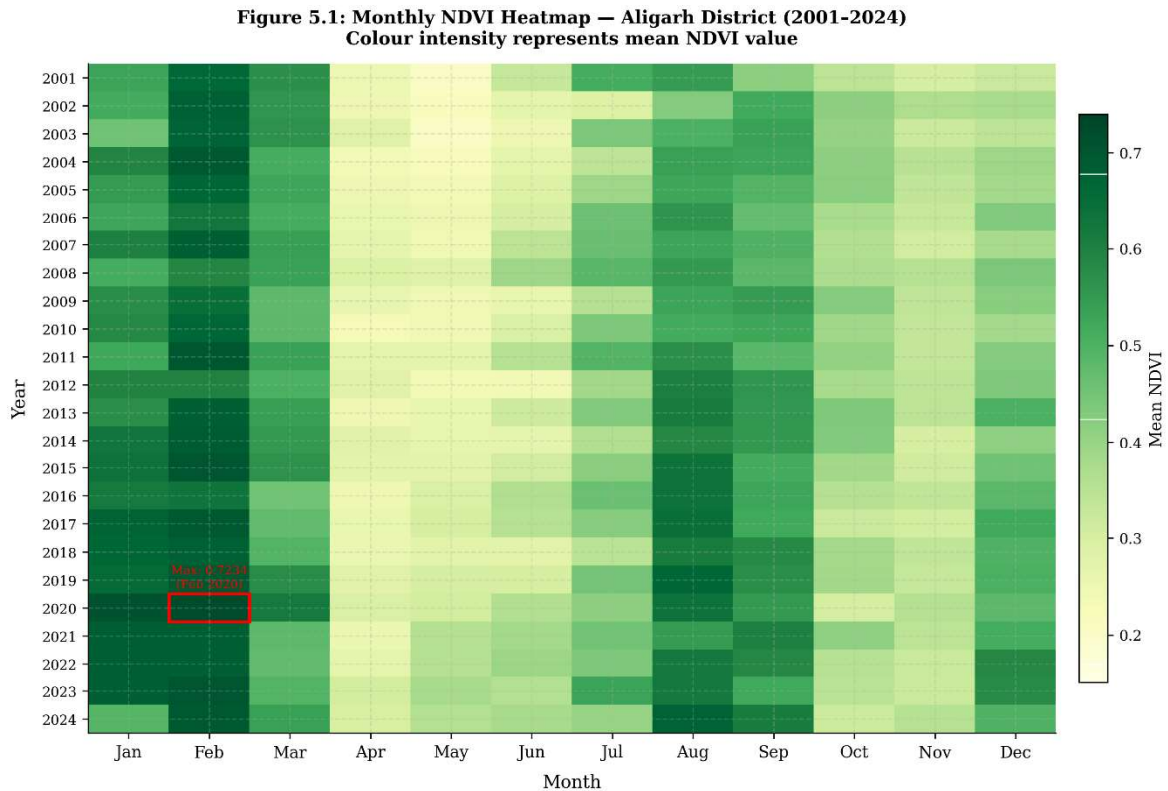
Positive = model underestimated | Negative = model overestimated

Table 4.10: Month-by-Month Validation of Monthly NDVI Forecast (2025)

Month	Actual NDVI	Forecast NDVI	Abs. Error	% Error	Accuracy
January	0.6373	0.6183	0.0190	2.98%	Satisfactory
February	0.6919	0.6641	0.0278	4.02%	Satisfactory
March	0.4505	0.5097	0.0592	13.14%	Moderate
April	0.2715	0.2644	0.0071	2.61%	Very Good
May	0.3876	0.2760	0.1116	28.79%	Weak
June	0.4095	0.3239	0.0856	20.90%	Moderate
July	0.4126	0.4612	0.0486	11.78%	Satisfactory
August	0.6690	0.5729	0.0961	14.37%	Moderate
September	0.5533	0.5302	0.0231	4.18%	Very Good
October	0.3806	0.3746	0.0060	1.58%	Very Good
November	0.3597	0.3373	0.0224	6.22%	Satisfactory
December	0.5423	0.4965	0.0458	8.44%	Satisfactory

seasonal transitions. The median and mean of monthly mean NDVI are closely aligned (0.4364 vs 0.4399), indicating approximate distributional symmetry across the dataset.

5.3.2 Monthly NDVI Heatmap



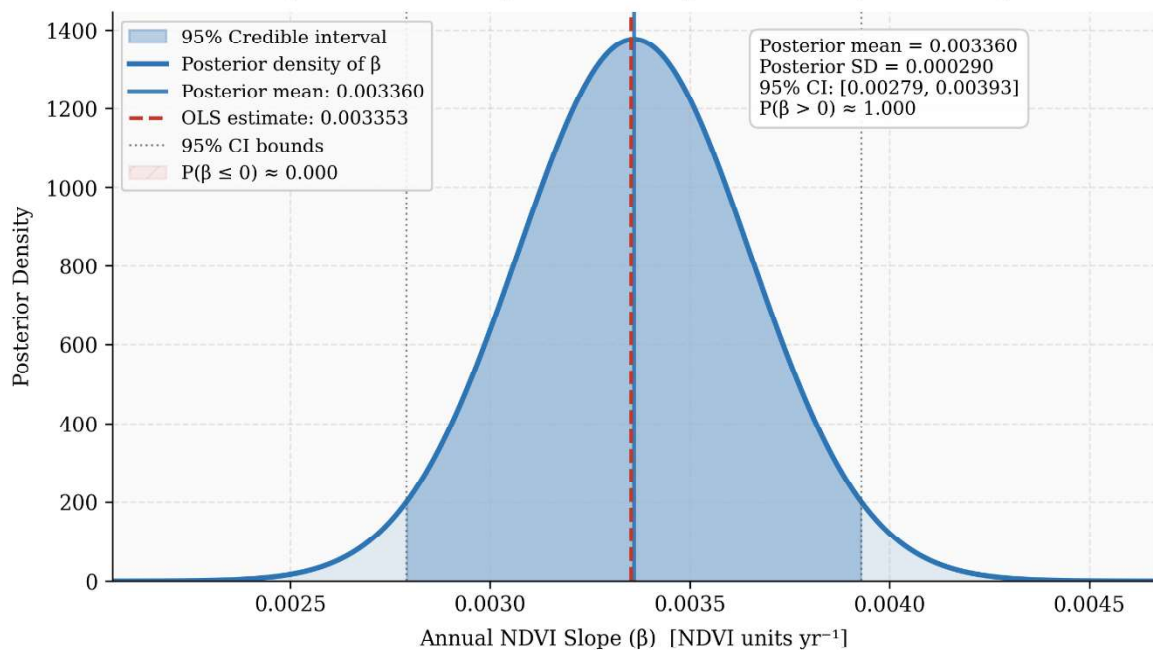
Monthly NDVI Heatmap for Aligarh District (2001–2024). Each cell represents the mean NDVI for a given month–year combination. Darker green = denser vegetation. The stable February column (Rabi peak) and April–May trough are visible across all years. A gradual lightening towards more recent rows reflects the long-term upward trend.

Figure 5.1: Monthly NDVI Heatmap for Aligarh District (2001–2024)

Figure 5.1 presents a two-dimensional heatmap of monthly mean NDVI organised by month (x-axis) and year (y-axis), with colour intensity representing the NDVI value. The heatmap provides a compact visual summary of both the seasonal structure and the long-term trend simultaneously. The warm-coloured cells (high NDVI) in the February column are consistently present across all 24 years, confirming the stable Rabi season peak. The cool-coloured cells (low NDVI) in the April–May column are equally consistent across all years, reflecting the regular inter-seasonal trough. A gradual shift towards warmer colours in the

5.4.6 Posterior Density of the Slope Parameter

Figure 5.3: Posterior Probability Distribution of the Annual NDVI Slope Parameter Bayesian Linear Regression — Aligarh District (2001–2024)



Posterior Probability Distribution of the Annual NDVI Slope Parameter (β). Distribution centred at 0.003360 (SD = 0.000290). Shaded region = 95% credible interval [0.00279, 0.00393]. Red line = OLS estimate (0.003353). Entire distribution is positive: $P(\beta > 0) \approx 1.000$.

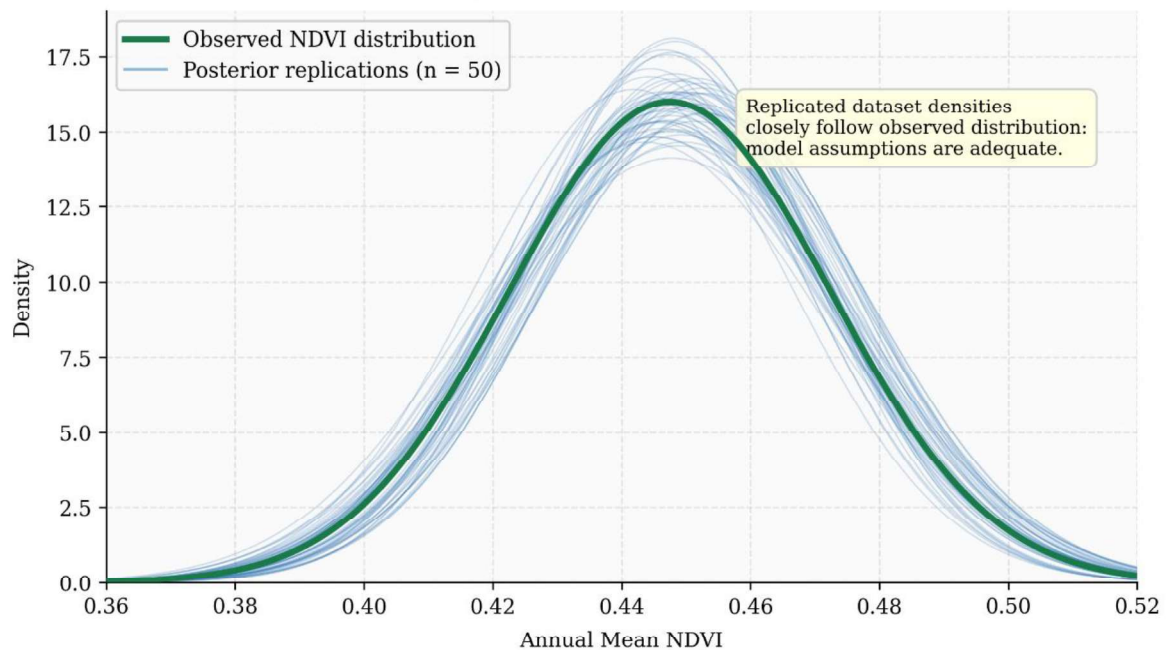
Figure 5.3: Posterior Probability Distribution of the Annual NDVI Slope Parameter (β)

Figure 5.3 displays the posterior density of the slope parameter. The distribution is approximately normal in shape, centered at 0.003360, and entirely located in the positive domain. The 95% credible interval boundaries are visible as the limits within which the bulk of the posterior density is concentrated. The absence of any posterior mass at or below zero provides a visual confirmation of the near-certain positive direction of the NDVI trend. The narrow width of the posterior distribution — spanning approximately 0.0012 NDVI units — reflects the high precision of the slope estimate, which in turn results from the strong and consistent signal in the 24 years of annual NDVI data.

Posterior density visualisation and credible interval assessment were performed using density plotting functions from the bayesplot package in R. Complete implementation details are provided in Appendix A, Section A.6.

5.4.7 Posterior Predictive Check

**Figure 5.4: Posterior Predictive Check — Bayesian Linear Regression Model
Aligarh District (2001-2024)**



Posterior Predictive Check for the Bayesian Linear Regression Model. Blue curves = density of 50 synthetic NDVI datasets drawn from the posterior predictive distribution; green curve = observed NDVI density. Close agreement confirms model adequacy.

Figure 5.4: Posterior Predictive Check for the Bayesian Linear Regression Model

Figure 5.4 presents the posterior predictive check, in which 50 synthetic NDVI datasets are generated from the posterior predictive distribution and their densities overlaid against the observed NDVI density. The replicated dataset density curves closely follow the shape and range of the observed NDVI density, confirming that the Bayesian model can generate data with characteristics consistent with those of the original observations. No systematic discrepancy between the simulated and observed densities is apparent. This result confirms that the model's assumptions are appropriate for the yearly NDVI data and that the posterior estimates are reliable.

Posterior predictive assessment was conducted using posterior predictive checking functions from the `bayesplot` package in R to compare replicated and observed NDVI distributions. Complete implementation details are provided in Appendix A, Section A.6.